Solving the SDE for OU process

$$dX_t = \gamma(m - X_t)dt + \sigma dB_t \qquad t>0 \qquad (*)$$

Let $Y_t = X_t - m$ distance from X_t to E $\rightarrow Y_{t\to\infty} = 0$

$$\rightarrow dY_t = dX_t - dm = dX_t$$

$$\xrightarrow{(*)} dY_t = -\gamma Y_t dt + \sigma dB_t$$

multiply by $e^{\gamma t}$

$$e^{\gamma t} dY_t = -e^{\gamma t}\gamma Y_t dt + e^{\gamma t}\sigma dB_t$$

$$e^{\gamma t} dY_t + e^{\gamma t}\gamma Y_t dt = e^{\gamma t}\sigma dB_t$$

$$e^{\gamma t} dY_t + Y_t d(e^{\gamma t}) = e^{\gamma t}\sigma dB_t$$

$$d(e^{\gamma t} Y_t) = e^{\gamma t}\sigma dB_t$$

intergrating from 0 to t

$$\int_0^t d(e^{\gamma s} Y_s) = \int_0^t e^{\gamma s}\sigma dB_s$$

$$e^{\gamma t}Y_t - e^{\gamma 0}Y_0 = \sigma \int_0^t e^{\gamma s} dB_s$$

$$e^{\gamma t}Y_t - Y_0 = \sigma \int_0^t e^{\gamma s} dB_s$$

$$e^{\gamma t}Y_t = Y_0 + \sigma \int_0^t e^{\gamma s} dB_s$$

multiply back by $e^{-\gamma t}$

$$Y_t = e^{-\gamma t}Y_0 + e^{-\gamma t}\sigma \int_0^t e^{\gamma s} dB_s$$

$$Y_t = e^{-\gamma t}Y_0 + \sigma \int_0^t e^{-\gamma (t-s)} dB_s$$

Since $Y_t = X_t - m \rightarrow Y_0 = X_0 - m$

$$X_t - m = (X_0 - m)e^{-\gamma t} + \sigma \int_0^t e^{-\gamma (t-s)} dB_s$$

$$\boxed{X_t = m + (X_0 - m)e^{-\gamma t} + \sigma \int_0^t e^{-\gamma (t-s)} dB_s}$$